

DisasterClaw: A task-centric benchmark for active observation in embodied UAV inspection

Anonymous authors

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Abstract

Aerial agents can actively acquire new observations to resolve uncertainty, yet a new view can improve or degrade the final task answer. Existing evaluations often assess navigation or perception without isolating whether a new observation improves the final answer. To address this gap, we introduce DisasterClaw, a georeferenced benchmark and replay platform for evaluating task-conditioned observation decisions in disaster assessment. DisasterClaw keeps the question, geographic region, and reference answer fixed while allowing vehicle actions to change the observed image footprint. Its traceable evaluation protocol connects observations, predicted evidence, actions, and answers, and uses matched fixed-view controls to distinguish the value of new visual evidence from additional answer calls. The benchmark comprises 160 human-reviewed questions across 44 regions and three disaster events. Text-only and permitted-metadata controls achieve accuracies of 28.75% and 36.25%, respectively, compared with 72.29% and 75.00% for image-aware policies. Across 161 recheck transitions, changed-view accuracy reaches 59.63%, versus 50.31% for matched fixed-view calls, yielding a gain of 9.32 percentage points with a 95% region-clustered interval of [-2.00, 21.74]. These transitions produce 36 corrections and 21 harms, highlighting both the potential benefits and risks of re-observation. An additional offline case study applies a single-temporal adaptation of the controller to 409 eligible image pairs from MESSI, using existing YOLO and SegFormer models. The controller requests 20 lower-altitude observations, increasing the number of correct answers from 369 to 372; on these triggered cases, lower-altitude observations yield 15 correct answers, compared with 12 under an equal-budget repeat-high control. With the same 20-recheck budget, 20 fixed random allocations average 369.2 correct answers (range 367–371), compared with 372 for the entropy trigger on this cohort. Together, these experiments provide a reproducible basis for studying when additional observations help aerial agents, while leaving consistent cross-scene gains, flight-cost effectiveness, and physical flight effectiveness unestablished. Project resources will be available at <https://github.com/emmmmmmmnbbb/DisasterClaw>.

Keywords: Embodied aerial agents, Active perception, Disaster assessment, Observation value, Benchmarking

1. Introduction

Embodied AI for aerospace vehicles links onboard sensing, task reasoning, and executable actions under vehicle and mission constraints. In aerial disaster inspection, an agent may be asked whether a building is damaged, how many damaged buildings lie within a region, or where the nearest damaged building is located. Descending toward a building may improve local detail while narrowing the visible footprint and excluding evidence needed for a regional assessment. The value of an additional observation therefore depends on the question being answered and the maneuver it requires.

Embodied multimodal models connect visual and state observations to task reasoning, while language-guided navigation and vision-language-action systems ground plans in executable skills and actions [1, 2, 3, 4]. Aerial navigation and embodied benchmarks evaluate action-conditioned observations and mission behavior; remote-sensing agents support disaster interpretation and struc-

tured tool use [5, 6, 7, 8, 9, 10, 11, 12]. These capabilities motivate an evaluation question that is narrower but consequential: how does a vehicle’s decision to acquire a new view affect the answer to a task whose geographic scope and reference remain fixed? Answering it requires a controlled observation change and a trace linking the action to geographic evidence and the answer call. A standardized episode interface can make this comparison repeatable by distinguishing a requested action from an executed movement and separating changed-view effects from another answer opportunity.

We present DisasterClaw, an imagery-based platform and task benchmark for this evaluation problem. Paired xBD tiles, building annotations, task regions, and simulated vehicle states share a common geographic reference. Annotation-derived answers remain fixed as the agent searches or changes its observation footprint. Four task families impose different requirements for local detail, regional coverage, and geographic reference. An operator

console supports scene inspection, and a headless runner executes the quantitative evaluation.

The evaluation connects observation changes to building predictions and task answers. A camera-ladder diagnostic on the development set measures how a narrower footprint affects building localization and damage classification, including corrections and harms. On the frozen final set of 160 human-reviewed questions from 44 regions in three events, shortcut controls assess performance using only question text or permitted runtime metadata. A fixed-view matched-call diagnostic then branches at every executed T1 recheck: one branch reruns perception and answering on the previous image, and the other executes the logged recheck. Matching the answer opportunity helps distinguish the effect of changed observations at individual transitions. A separate MESSI case compares real UAV images acquired at two heights with repeat-view and digital-crop controls. We then inventory the full 2,525-image release and select 409 eligible two-height vegetation regions under the fixed protocol. On these pairs, we run a single-temporal adaptation of DisasterClaw with its existing YOLO and SegFormer models; the controller sees only the high view until its recheck gate authorizes the registered lower view. This measures adapted offline re-observation on real UAV imagery, alongside the platform’s primary xBD evaluation.

The contributions are:

1. a georeferenced, ROI-scoped task benchmark in which the reference answer stays fixed as the observation changes, with an explicit boundary between runtime task cues and scoring labels;
2. an inspection interface that records requested and executed actions, rendered views, predicted evidence, answer calls, and terminal outcomes within the same episode; and
3. an empirical characterization of observation value through shortcut controls, fixed-view matched calls, and corrections and harms by task, using a fixed task-conditioned policy as a reference implementation.

The platform renders existing orthographic imagery under simplified motion. We position it as an evaluation component for embodied aerial agents: it measures task-conditioned sensing decisions and answer consequences, while leaving vehicle dynamics, onboard sensing, and physical flight to future evaluation. The findings apply to this controlled observation setting; they do not establish flight effectiveness.

2. Related work

2.1. Aerial interaction and embodied evaluation

AerialVLN, CityNav, and OpenFly emphasize language-conditioned aerial navigation, while AeroVerse and BEDI broaden evaluation toward perception, spatial reasoning, planning, motion, and perception–decision–action loops [5, 6, 7, 8, 9]. CityNavAgent uses hierarchical semantic planning and trajectory memory for urban aerial

navigation, and the ICLR 2025 UAV platform and benchmark work combines realistic flight simulation with target-oriented navigation tasks [13, 14]. AerialClaw, a recent preprint, connects language-specified missions to executable UAV skills, runtime feedback, and simulation [15]. Together, these systems evaluate action-conditioned observations and broader aerial mission behavior. Aerial simulators such as AirSim and Flightmare couple sensing to vehicle dynamics and 3D rendering [16, 17]. DisasterClaw addresses a complementary question: whether a particular observation change helps a fixed geographic task, with the answer-level effect recorded separately from an extra inference call.

2.2. Disaster and remote-sensing agents

xBD supplies paired disaster imagery and building-damage annotations [18]. RescueADI, ThinkGeo, and DisasterM3 study tool use or multimodal disaster interpretation from remote-sensing records [11, 12, 19]. These resources motivate disaster-specific questions. Evaluating an answer from a static image, however, does not measure the value of an agent’s decision to observe again. DisasterClaw binds question scope, vehicle state, rendered observation, and terminal answer in one record. This binding also exposes a failure mode: a view that improves one building prediction can remove evidence required by a count or spatial question.

Table 1 summarizes the evaluation capabilities most relevant to this distinction. The comparison focuses on evaluation capabilities; it does not rank the overall scope of the benchmarks. Among the compared benchmarks, DisasterClaw explicitly standardizes a fixed annotation-derived geographic task and answer under view change, geographic evidence traces across observations, and a matched fixed-view answer call. These features define its contribution within a controlled observation setting.

2.3. Task value of an observation

Active perception treats sensing as a task-directed action, from foundational active-vision formulations to learned sequential localization policies [20, 21, 22]. In an embodied system, uncertainty or action feasibility can inform a sensing decision, but predictive uncertainty alone does not measure whether the subsequent task answer is correct [23, 24, 25]. We therefore report localization, answer corrections and harms, executed rechecks, and terminal accuracy separately. The task-conditioned rule in this paper is a fixed heuristic for descending subject to predicted ROI coverage and motion cost. Complete-policy comparisons measure outcomes at the realized operating points. Fixed-view matched calls at executed recheck transitions control for an additional answer opportunity; the MESSI repeat-high arm further compares a real new view with a second inference on the same view.

Table 1: Capability-oriented comparison of benchmarks most relevant to task-dependent observation value.

Benchmark	Primary task	Fixed geographic				
		Active view change	Disaster-specific	task/answer under view change	Answer after active sensing	Explicit observation-value isolation
AerialVLN	Aerial vision-language navigation	✓	—	—	—	—
OpenFly	Large-scale aerial vision-language navigation	✓	—	—	—	—
BEDI	UAV embodied-agent evaluation	✓	△	—	△	—
CityEQA	Embodied question answering	✓	—	△	✓	—
RescueADI	Disaster remote-sensing agent	—	✓	—	△	—
ThinkGeo	Tool-augmented remote-sensing agent	△	△	—	△	—
DisasterM3	Disaster vision-language benchmark	—	✓	—	—	—
DisasterClaw	Disaster inspection and observation-value measurement	✓	✓	✓	✓	✓

“—” means that the capability is not the primary evaluation contract reported by the work, rather than that it is technically impossible. △ denotes a related but non-equivalent capability. DisasterClaw changes controlled orthographic image footprints rather than physical 3D viewpoints.

3. The DisasterClaw platform

3.1. From geographic records to inspection episodes

DisasterClaw links each task, observation, and answer to a common geographic scene (Fig. 1). Paired xBD images are positioned with pixel-to-geographic affine transforms. Building polygons and damage labels define the reference task; a vehicle state and observation window determine what the agent can see. The operator console and headless runner share the same scene and controller modules. All reported quantitative results are produced by the headless runner.

The active disaster tile is embedded at its recorded map footprint. Geographic coordinates locate the task ROI, marked building when applicable, vehicle, and rendered window. Disaster imagery supplies the annotated task evidence; a basemap can supply context outside available tiles but contributes no damage ground truth. The task answer remains defined on the ROI when search or recheck changes the observation window. The experiments use this fixed-answer, variable-view design to evaluate observation value.

3.2. Controlled observation and action

The nadir renderer uses a 60° horizontal field of view and a 1024-pixel output width. At height h , the ground-footprint width L and nominal ground sampling distance g are

$$L(h) = 2h \tan(\theta/2), \quad g(h) = \frac{L(h)}{W}. \quad (1)$$

The assumed source scale is 0.5 m/pixel, so a 1024-pixel tile defines a nominal 512 m width. Footprints of three, two, and one tile widths give cruise, intermediate, and floor states at approximately 1330.2, 886.8, and 443.4 m, with nominal GSDs of 1.5, 1.0, and 0.5 m/pixel (Fig. 2). Geographic placement comes from the recorded affine transforms; the 0.5 m/pixel value defines the camera ladder without assuming an independently verified physical scale for every source tile.

For paired damage perception, the post-disaster image is sampled at the current footprint and output size; the pre-disaster window is retained at the assumed source scale

before both channels enter the detector input geometry. This asymmetric preprocessing means that the camera-ladder response combines changes in footprint and context with resampling and detector-scale effects. It cannot be attributed solely to additional physical image detail.

The vehicle model stores geographic position, height, heading, and task state. Search can move the vehicle and produce another rendered observation. A dedicated recheck may center on a target and descend, subject to an execution and budget check. The episode log distinguishes requested actions from executed movements, observation identifiers, predicted objects, answer calls, and the final normalized answer. An unchanged-view answer mode supports diagnostic evaluation and is separate from the frozen policy comparison reported here. Motion is simplified position interpolation, and rendering existing orthographic imagery does not create new viewpoints or acquisition times.

4. Task benchmark construction

4.1. Fixed task scope and reference answers

Each question specifies a source tile, a geographic region of interest (ROI), an initial vehicle state, an answer vocabulary, and an annotation-derived reference answer. A building belongs to the reference ROI when its centroid lies inside the ROI bounds. Let B_R be these buildings and $D_R \subseteq B_R$ the buildings labelled damaged. Minor, major, and destroyed xBD labels map to the binary damaged class; unclassified records do not define a binary answer. The reference answer is computed before a policy runs and is unchanged by later crops or movements.

The four task families capture different aspects of the trade-off between local detail and regional coverage. Presence asks whether $|D_R| > 0$: a single convincing positive may suffice, whereas a negative answer requires evidence across the region. Damage asks for the binary state of a designated building, whose location is a permitted task cue. Count maps $|D_R|$ to 0, 1, 2, or 3+ and requires aggregation across the ROI. Spatial asks for the eight-sector bearing from the ROI center to the nearest damaged building and requires a nonempty D_R . Reference answers are derived

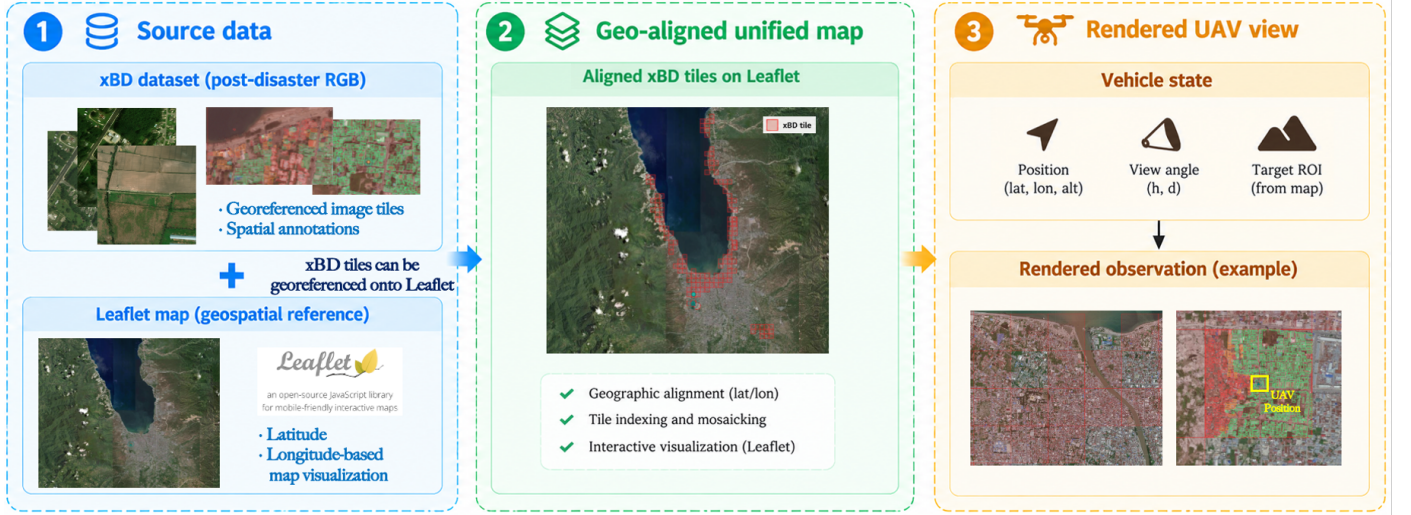


Figure 1: Dataset and scene construction. Georeferenced xBD imagery and spatial annotations are aligned with a Leaflet map, indexed into a unified geographic scene, and rendered according to vehicle state and the target ROI. The basemap supplies geographic context but no damage ground truth.

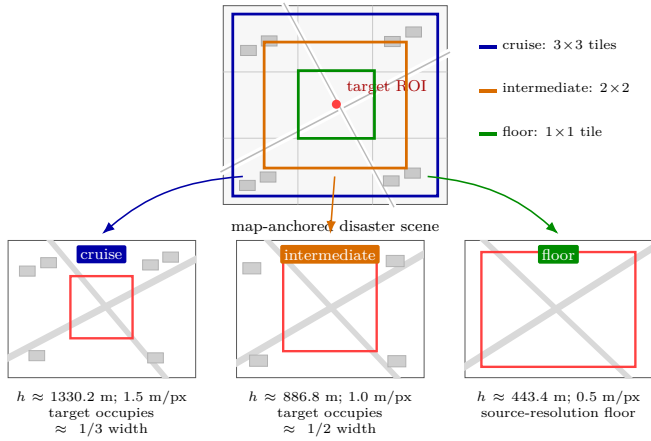


Figure 2: Camera-ladder abstraction. Each state renders 1024×1024 pixels from the same orthographic source; descending narrows the footprint and cannot reveal detail below the assumed source resolution.

from these annotation and geometry rules, independently of language-model generation.

Figure 3 illustrates how platform inputs are converted into task outputs. Spatial localization associates predicted damaged buildings with geographic position before computing a relative direction; damage counting aggregates building-level predictions over the visible region; and an active inspection instruction expands into navigation, descent, observation, and reporting actions. These platform-run examples illustrate the interface and are separate from the frozen 160-question evaluation set.

4.2. Final-set composition and input boundary

The frozen set has 160 questions over 44 ROIs: 48 in Hurricane Michael, 51 in the Moore tornado, and 61 in Nepal flooding. Table 2 reports the answer counts. The count task is concentrated in 0 and 3+, with only one 1 and no 2; this class imbalance prevents count accuracy alone from establishing performance across all four categories. Spatial answers are more dispersed, but that task also requires finding the correct nearest damaged building. Questions from the same ROI share image context and therefore do not represent independent scenes.

Table 2: Frozen question and answer distribution. Counts refer to distinct questions, before policy runs or generation repeats.

Task	n	Reference answers (count)
Presence	44	yes 30; no 14
Damage	44	damaged 29; no damage 15
Count	43	0: 14; 1: 1; 2: 0; 3+: 28
Spatial	29	N 3; NE 4; E 3; SE 4; S 3; SW 3; W 4; NW 5

The source task record separates runtime inputs from scoring information. The controller receives the question,

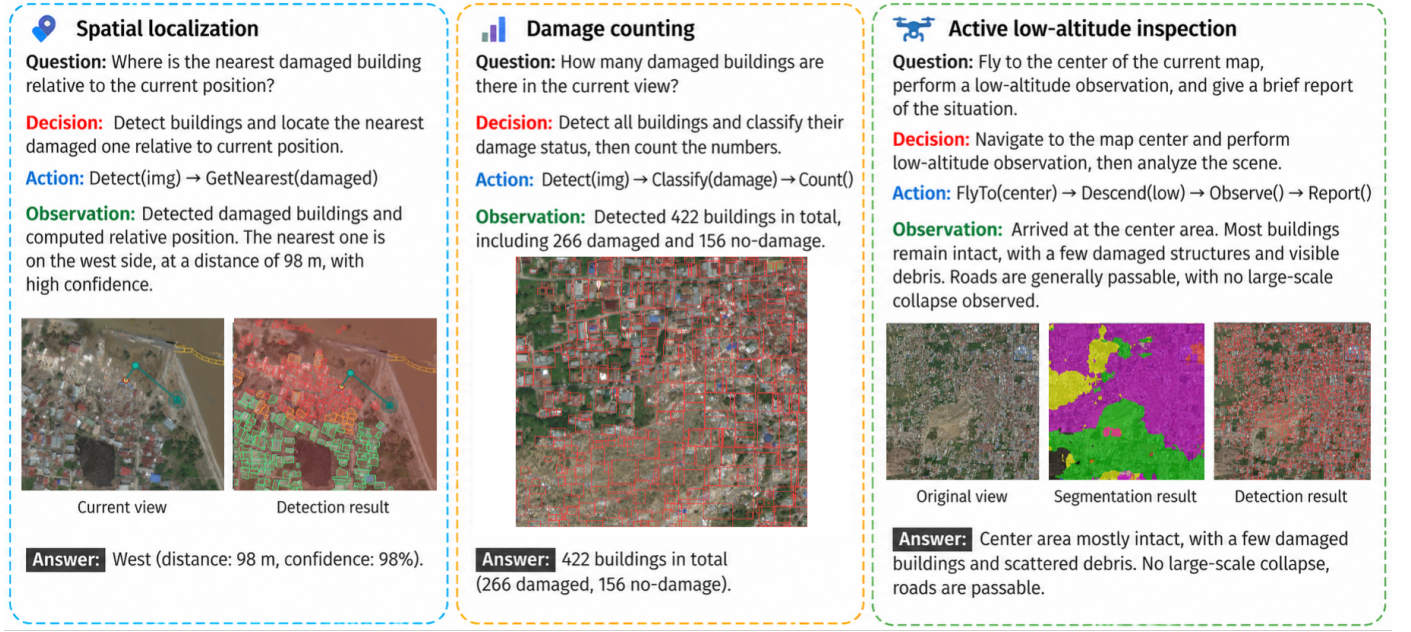


Figure 3: Representative platform-run input-output traces. The examples show spatial localization from mapped detections, aggregation for damage counting, and an active low-altitude inspection sequence. They illustrate the interface and action vocabulary; quantitative results use the separately frozen benchmark and headless runner.

answer choices, ROI bounds, initial state, and a marked reference ID and coordinate only for a damage question. It receives detector-predicted boxes, probabilities, and geographic projections after observation. The reference answer, xBD damage labels and polygons, and the true nearest-damaged-building coordinate for a spatial question are restricted to benchmark construction and scoring. For spatial questions, the agent must select a target from predicted evidence because the ground-truth target coordinate is withheld.

Eligible records require usable post-disaster imagery, a geographic footprint, and building labels; pre-disaster imagery supplies the archival damage reference. The generator excludes consumed development tiles, checks answer vocabulary, geometry, duplicate records, and split membership, and retains ambiguity flags for review. The final questions underwent human review before the set was frozen. These checks document reference construction and the input boundary. Visual answerability and potential shortcuts require separate evaluation. Section 7 therefore evaluates text-only and permitted-metadata controls on the same frozen set. The comparison assesses whether the complete image-aware pipeline outperforms controls using the permitted non-image inputs. Because the controls also use a modality-specific prompt and omit the hybrid rule path, the contrast does not isolate the causal effect of image access alone.

5. Task-conditioned observation policy

5.1. Episode state and evidence memory

An episode begins with a question q , geographic ROI, initial state $s_0 = (x_0, y_0, h_0)$, and fixed reference answer y_q . At step t , the renderer produces $I_t = \mathcal{R}(\mathcal{M}, s_t)$; ChangeOS supplies predicted building locations and binary damage probabilities, and the hybrid answer pipeline using Qwen2.5-VL-7B-Instruct proposes an answer. The controller can answer, abstain, search, or request a dedicated recheck. The normalized terminal answer is scored against y_q , with abstentions counted as incorrect for the primary endpoint. Every policy uses the same frozen ChangeOS checkpoint and answer path.

Predicted objects are projected into geographic coordinates. For non-damage tasks, sightings are associated when their predicted centers are within 6 m or their geographic boxes have intersection-over-union at least 0.2; damage sightings use the public marked-building reference ID. Each track retains observation IDs and averages binary probabilities with inverse-GSD weights. Count and spatial answers use tracks detected in the current observation, so a stale historical detection cannot independently supply a current nearest target. The memory may still merge nearby buildings or miss objects, even when geometric ROI coverage is high.

5.2. Task-conditioned recheck rule

For a candidate recheck, T1_TASK uses

$$U_q = u_q(1 - h_{t+1}/h_t) - 0.05 \hat{T}_{\text{move}}/60 - (1 - \hat{C}_{\text{ROI}}), \quad (2)$$

where $u_q \in [0, 1]$ is a task-specific uncertainty score, $\hat{T}_{\text{move}}$ is configured-speed motion time in seconds, and $\hat{C}_{\text{ROI}}$ is predicted coverage after the move. A candidate executes only when $u_q \geq 0.5$, $U_q \geq 0.05$, and, for count or spatial questions, predicted ROI coverage is at least 0.98. Only descent contributes positively to the utility. The rule therefore triggers descent subject to task uncertainty and coverage constraints; it neither assigns value to same-height viewpoint changes nor estimates calibrated accuracy gains.

The uncertainty score is defined separately for each task. For damage, it is normalized binary entropy for the marked predicted building; a visible marked location with no matched detection receives $u_q = 1$. For count, a capped Poisson-binomial distribution from detected buildings’ damage probabilities gives entropy over 0, 1, 2, 3+. For a matched spatial candidate, the score is the maximum of its binary entropy and a GSD-dependent estimate of proximity to an eight-sector bearing boundary. Presence is answered from the wide view when possible; missing evidence enters search rather than a local recheck. Count uncertainty is conditional on detected objects and does not account for missed buildings. The spatial score uses a selected predicted target and does not fully represent uncertainty about which damaged building is nearest.

Damage rechecks may move toward the marked building by at most 40 m horizontally and descend. Count and spatial rechecks keep the geographic center and descend only while the task ROI remains covered. Each active policy has a cap of two dedicated rechecks. The stored policy identifier A0_HOLD is referred to here as *A0_NO_RECHECK*: it has a zero dedicated-recheck cap but, like the other policies, may use up to six search steps. Search may move and render a new image. Thus *A0_NO_RECHECK* is not a fixed-view control, and the reported recheck count is not total movement, rendering, or model-call cost.

The primary comparators are A1_RANDOM (a seeded 0.5 recheck trigger), A2_ALWAYS (request a feasible recheck), and A3U_RAW_ENTROPY (uncalibrated binary-entropy trigger). A3_ENTROPY and A4_CONFORMAL are supplemental policies. All seven configurations share the same question set and search cap, but their executed actions and subsequent answer-call counts may differ. This comparison therefore measures whole-policy outcomes at their realized operating points.

6. Evaluation protocol

The evaluation links changes in observation to building predictions and task answers. The development camera-ladder diagnostic compares cruise, intermediate, and floor observations for the same 7,238 annotated buildings in 43 ROIs. Its fixed-denominator measure counts unmatched ground-truth buildings as incorrect. Classification accuracy among matched buildings describes a selected subset and is not used as the task endpoint. This diagnostic helps interpret whether the rendered view changes available predicted evidence.

The final online set is separate from the 160-question development set used to select policy rules. Seven frozen policies each answer the same 160 final questions under three generation repeats, yielding $7 \times 160 \times 3 = 3,360$ episodes. The repeats vary seeded generation conditions for the same questions; they do not increase the number of independent questions, ROIs, or events. The primary endpoint is terminal question accuracy, with abstentions counted incorrect. Executed dedicated rechecks, within-policy answer transitions, task type, and event are descriptive companion measures.

We audited spatial separation beyond distinct tile IDs. The 43 development ROIs and 44 final ROIs share no tile ID or xBD building UID (0 shared among 7,238 and 4,963 eligible ROI buildings), but 9/1,892 cross-set ROI pairs intersect; the maximum ROI IoU is 0.461 and the maximum fraction of the smaller ROI covered by the other is 0.631. Comparing 129 centered development camera-ladder footprints with all 802 recorded T1 final observation footprints across three repeats, 1,871/103,458 cross-set pairs intersect. The maximum footprint IoU is 0.882, and one smaller footprint lies entirely inside a larger one (overlap fraction 1.000). These measurements identify overlapping geographic windows but do not establish whether both sets contain the same xBD pixels or physical buildings. Different tiles may assign different UIDs to the same physical structure. Thus question and tile disjointness does not establish spatial independence.

We add two controls on the frozen final set. First, the task-majority baseline predicts one answer per task type using a question/ROI-disjoint development set. S1_LANGUAGE receives question text, choices, and task type; S2_METADATA additionally receives runtime-visible ROI bounds/tile, initial pose, and, for damage questions, the permitted marked-building reference and coordinate. Both exclude images, detector outputs, annotation labels and polygons, the reference answer, and the spatial true-nearest target. S1 and S2 use the same Qwen checkpoint, answer vocabulary, generation settings, seed schedule, and exact-match scoring as the main answer model, but a modality-appropriate text-only JSON prompt and no hybrid rule path.

Second, a fixed-view matched-call diagnostic is run at every one of the 161 executed T1 dedicated rechecks. Immediately before the live recheck, the fixed branch clones the semantic map and evidence memory, remains at the old pose, reruns perception on the old view, and calls the ordinary hybrid answer path. The changed branch executes the logged action, perceives the new view, and calls that same answer path. The two branches share question, repeat, next-step seed, model, and generation settings; their prompts differ only through the resulting predicted evidence. A strict gate requires a byte-identical fixed-branch marked image and exactly one fixed and changed model call with the shared seed. The diagnostic compares paired candidate answers immediately after the transition, without following both branches to episode termination. The

fixed-branch answer is never used by the T1 controller.

Four T1_TASK contrasts were specified before the final run: against A0_NO_RECHECK (internal identifier A0_HOLD), A1_RANDOM, A2_ALWAYS, and A3U_RAW_ENTROPY. Comparisons pair the same question and repeat. The 5,000-draw bootstrap resamples ROI clusters, retaining their questions and repeats, and reports Bonferroni-adjusted 98.75% intervals for the four contrasts. The intervals characterize uncertainty across the 44 observed ROI clusters. The three-event sample does not support inference about transfer to unseen disasters. Task and event breakdowns are descriptive, not additional adjusted tests.

For the matched diagnostic, correctness pairs and answer disagreement are summarized overall and by task, with a 5,000-draw ROI-cluster bootstrap 95% interval for the changed-minus-fixed difference. Shortcut results report overall and task-wise accuracy; image-aware A0_NO_RECHECK and T1 gaps to S1/S2 are paired by question and repeat with the same ROI-cluster bootstrap. These controls assess visual dependence and observation value and are not used for additional policy selection.

The primary cost count records executed dedicated rechecks. Search can also move and render, and policies can make different numbers of later Qwen calls; total movement, rendering, perception, and generation costs are therefore not summarized by the recheck column. A supplementary archive audit reports observation and generation-attempt counts, search actions, state-derived movement on complete trajectories, and episode wall time (Table 8). Episode wall time is measured software runtime, not flight duration. The final protocol identifies the frozen task, model, prompt, and run configurations; the audit scope and planned release are summarized in Appendix A. A runnable single-episode scene and scoring replay is described in Appendix B.

7. Results and discussion

We first report the benchmark controls, matched-call observation effects, and complete-policy results. For real UAV imagery, the final adapted-controller evaluation is the 409-region YOLO/SegFormer replay in Section 7.7.1, including its equal-budget repeat-high control. The preceding MESSI experiments document exploratory development and fixed-observation diagnostics; they are not additional independent validations of that final controller.

7.1. Image-aware performance and shortcut controls

The final answer distribution yields a strong task-majority baseline: the development-fitted task-majority rule attains 57.50% (92/160). This result reflects the frozen answer distribution, particularly for count questions, without using visual evidence. The text-only S1_LANGUAGE control reaches 28.75% over 480 question-repeat instances, and the permitted-metadata S2_METADATA control reaches 36.25% (Table 3). Neither control receives imagery or detector evidence. In contrast, the image-aware

A0_NO_RECHECK and T1_TASK operating points reach 72.29% and 75.00%, respectively. Their paired gaps to S1 are +43.54 and +46.25 points, and to S2 are +36.04 and +38.75 points; all four 95% ROI-cluster intervals exclude zero. The complete image-aware pipelines therefore outperform the evaluated controls based on question wording and permitted non-image metadata. Because S1/S2 also use a modality-specific prompt and omit the hybrid rule path, the comparison measures a pipeline-level difference and does not isolate the causal effect of image access.

7.2. Observation changes under matched answer calls

The policy comparison alone cannot distinguish a new image from another answer opportunity. We therefore created paired branches at all 161 executed T1 rechecks, with each pair sharing its pre-recheck state. The fixed branch reran perception and answering on the previous view. The changed branch executed the original recheck before perception and answering. The strict validity gate accepted all 161 pairs, with no missing pair or execution error. The branch schedule reproduced the original recheck-step locations in 118 of 119 episodes; the one mismatch retained the same two rechecks but placed the second at a different step, reflecting re-runnable perception-path variation.

The changed-view branch produces correct candidate answers in 96/161 transitions (59.63%), versus 81/161 (50.31%) for the fixed-view matched call, a paired difference of +9.32 points (95% ROI-cluster interval $[-2.00, +21.74]$). The point estimate is accompanied by 36 fixed-wrong/changed-correct transitions and 21 fixed-correct/changed-wrong transitions; 44 pairs are wrong in both branches and 60 correct in both. Answers disagree in 42.24% of pairs. Thus the changed observation can alter the answer path beyond a repeated call on the old image, but the clustered interval does not establish a stable positive average effect in the sampled ROIs.

Excluding the one episode whose rerun placed its second recheck at step 2 rather than reference step 3 removes two wrong-wrong pairs. The remaining 159 pairs from 118 episodes have 81/159 fixed-view and 96/159 changed-view correct candidates, a +9.43-point paired difference (95% ROI-cluster interval $[-2.03, +22.00]$), with the same 36 help and 21 harm transitions. As a separate old-view repeat check, the pre-recheck and fixed-branch image bytes and normalized candidate answers agree in all 161/161 original forks (159/159 after exclusion). The two old-view calls use different step seeds and evidence prompts, however, so this is observed answer stability under identical image bytes rather than a controlled same-prompt, same-seed repeatability estimate.

Figure 4 grounds the aggregate transitions in two frozen T1 episodes. In the damage episode `hurricane-michael_00000435_post_disaster__damage_0001_9430`, successive observations at approximately 1330, 887, and 443 m increase the marked building’s predicted damage probability from 0.158 to 0.369 and 0.588. The answer consequently

Table 3: Shortcut controls and image-aware operating points on the frozen final set. S0 uses a development-fitted task majority and is deterministic ($n = 160$); all other rows use three aligned repeats ($n = 480$). S1 receives question text, choices, and task type; S2 additionally receives permitted runtime metadata.

Method	Image	Detector evidence	Runtime metadata	Presence	Damage	Count	Spatial	Overall
S0 task majority	no	no	task only	68.18	65.91	65.12	17.24	57.50
S1 language only	no	no	question only	31.82	34.09	32.56	10.34	28.75
S2 metadata only	no	no	permitted	31.82	58.33	32.56	14.94	36.25
A0_NO_RECHECK	yes	yes	permitted	93.18	66.67	75.97	43.68	72.29
T1_TASK	yes	yes	permitted	93.18	75.76	78.29	41.38	75.00

Table 4: Fixed-view matched-call diagnostic at executed T1 recheck transitions. Candidate-answer correctness is measured immediately after the branch, not at episode termination. Presence has no T1 dedicated recheck.

Task	n	Fixed	Changed	Help	Harm	Δ pp
Damage	123	46.34	58.54	27	12	+12.20
Count	18	83.33	100.00	3	0	+16.67
Spatial	20	45.00	30.00	6	9	-15.00
Overall	161	50.31	59.63	36	21	+9.32

changes from the incorrect “no damage” to the correct “damaged.” This example illustrates the potential benefit of a more local observation for a damage question.

In contrast, the spatial episode `hurricane-michael_0000306_post_disaster__spatial_0092_5452` shows the complementary risk. The wide observation selects a damaged building north of the ROI center and answers correctly. After descent, the evidence ranking selects a different building southwest of the center, changing a correct answer to an incorrect one. A further descent is rejected because predicted ROI coverage is 0.340, below the 0.98 gate. The failure involves a change in target identity under reduced regional context, rather than only bearing noise for a fixed target.

7.3. Building-level evidence and task-dependent outcomes

The camera-ladder diagnostic holds 7,238 annotated buildings in 43 development ROIs fixed while changing the rendered footprint. ChangeOS localization recall rises from 22.09% at cruise to 34.06% at intermediate and 56.52% at floor. With every unmatched ground-truth building counted as incorrect, binary building accuracy rises from 20.01% to 31.06% and 53.65%. Among matched buildings only, binary damage accuracy is 90.56%, 91.20%, and 94.92%, while matched-only macro-F1 is 0.8714, 0.8743, and 0.9232. The paired cruise-to-floor outcomes include 2,696 incorrect-to-correct changes and 261 correct-to-incorrect changes. The building-level diagnostic shows that rendering changes affect the predicted evidence available to the agent. Asymmetric pre-/post-image processing and changes in context prevent attributing these effects solely to GSD or physical descent.

Table 5 separates localization from conditional damage classification. Localization precision falls to 68.65% at

floor even as recall and matched-building damage accuracy rise; unmatched predictions increase to 43.44 per ROI. TP/FP/FN use a one-to-one nearest-centroid match within 12 m, so an unmatched prediction can reflect a duplicate, localization error, or missing annotation as well as a spurious building. The all-GT accuracy denominator includes missed buildings but does not penalize extra predictions; matched-only accuracy and macro-F1 exclude misses and unmatched predictions. The gains in damage accuracy and the increase in localization false positives thus describe different aspects of the detection pipeline.

The matched-call results show that task answers also respond differently to observation changes. Damage and count transitions have positive point differences, whereas spatial transitions have a negative point difference (Table 4). A more local view can improve the evidence for a marked building or a damaged-count decision, yet it can change the selected target for a nearest-building direction judgment. These task-wise estimates are descriptive because T1 selects the recheck subsets, which contain only 18 count and 20 spatial transitions.

7.4. Complete-policy performance

Table 6 reports the prespecified complete-policy runs. T1_TASK has the highest observed terminal accuracy, 360/480 question-repeat instances (75.00%), with 161 executed dedicated rechecks. A0_NO_RECHECK (the internal configuration identifier is A0_HOLD) forbids dedicated rechecks but permits search and answers 347/480 (72.29%). A2_ALWAYS makes 923 rechecks and answers 307/480 (63.96%). Active policies share a maximum of two dedicated rechecks. Their realized operating points differ in actions, model calls, and total cost, which are not matched in this comparison.

The 2.71-point T1_TASK advantage over A0_NO_RECHECK is an observed difference of 13 correct question-repeat instances, not 13 distinct questions. Its 98.75% multiplicity-adjusted interval includes zero (Table 7), as does the comparison with raw entropy. The positive interval against A2_ALWAYS supports a difference between these complete policies on the sampled ROIs; it does not show that T1’s individual changed observations caused that difference.

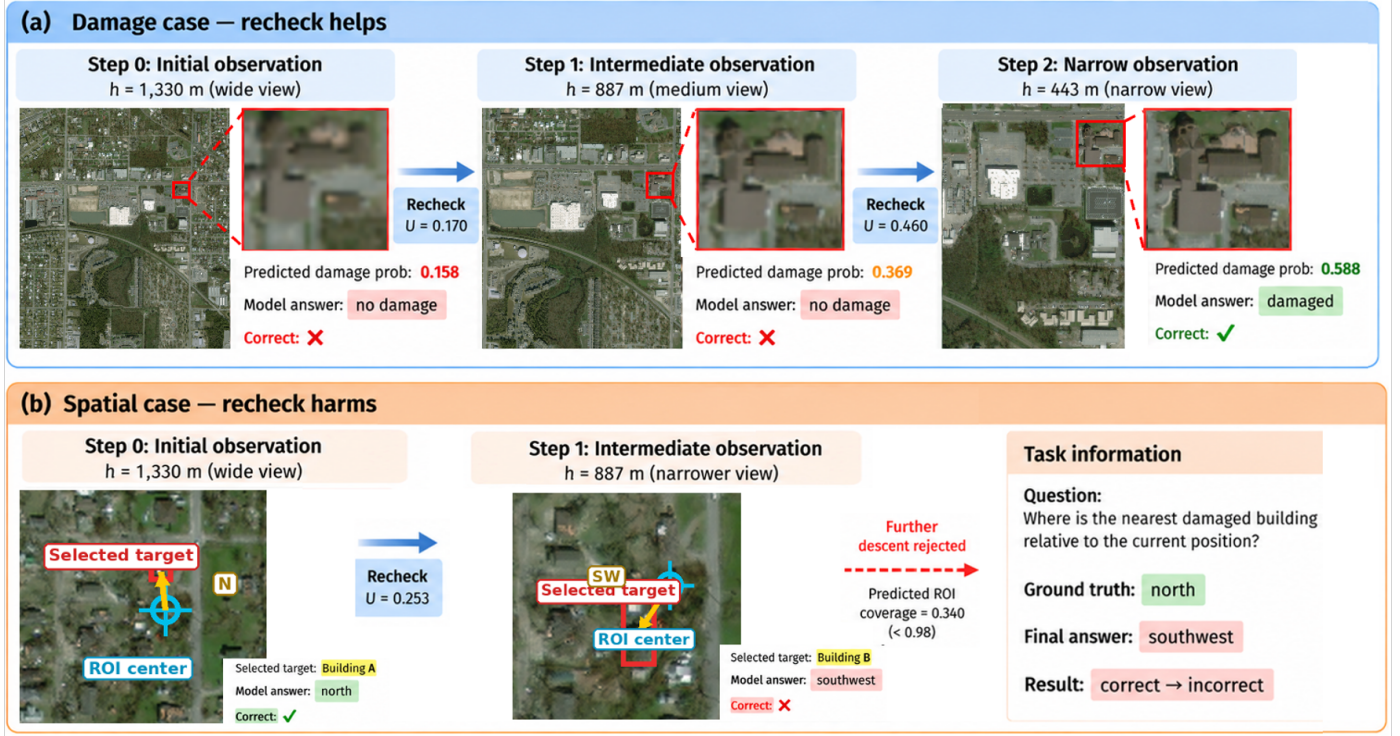


Figure 4: Frozen examples of recheck help and harm. (a) Narrower observations strengthen evidence for a marked building and correct a damage answer. (b) A spatial recheck changes the selected nearest damaged building from the correct northern target to a southwestern target; the coverage gate then rejects further descent. The examples expose the action–observation–evidence–answer chain behind the aggregate help and harm counts.

Table 5: ChangeOS localization and binary-damage audit on the same 43 development ROIs and 7,238 annotated buildings at each height. Loc. P/R are localization precision/recall; FP is an unmatched prediction, FN an unmatched GT building, and FP/ROI is FP/43. All-GT accuracy counts unmatched GT as wrong but does not count unmatched predictions; matched accuracy and macro-F1 condition on matched GT only.

View (height)	TP	FP	FN	Loc. P (%)	Loc. R (%)	FP/ROI	All-GT acc. (%)	Matched acc. (%)	Matched macro-F1
Cruise (1330 m)	1,599	502	5,639	76.11	22.09	11.67	20.01	90.56	0.8714
Intermediate (887 m)	2,465	677	4,773	78.45	34.06	15.74	31.06	91.20	0.8743
Floor (443 m)	4,091	1,868	3,147	68.65	56.52	43.44	53.65	94.92	0.9232

Table 6: Primary frozen policy operating points on the same 160 questions under three generation repeats. Rerechecks are executed dedicated actions and exclude search.

Policy	Correct / 480	Accuracy (%)	Rerechecks
A0_NO_RECHECK	347	72.29	0
A1_RANDOM	326	67.92	419
A2_ALWAYS	307	63.96	923
A3U_RAW_ENTROPY	339	70.63	626
T1_TASK	360	75.00	161

Table 7: Prespecified paired T1_TASK terminal-accuracy contrasts in percentage points. The 98.75% ROI-cluster intervals adjust for four comparisons.

Contrast	Difference	98.75% interval
T1 minus A0_NO_RECHECK	+2.71	[−2.04, +7.50]
T1 minus A1_RANDOM	+7.08	[+0.01, +14.69]
T1 minus A2_ALWAYS	+11.04	[+0.86, +21.38]
T1 minus A3U_RAW_ENTROPY	+4.38	[−4.33, +12.96]

At the terminal-episode level, presence is unchanged between T1 and A0_NO_RECHECK, damage and count are numerically higher for T1, and spatial accuracy is lower (Table 3). Performance also varies by event: T1 versus A0 is 54.86% versus 58.33% in Hurricane Michael, 84.31% versus 77.12% in Moore, and 83.06% versus 79.23% in Nepal. Together, the matched-call results and complete-policy comparisons characterize task-dependent corrections and harms. The interval for the A0 contrast and incomplete cost measurements limit conclusions about the overall effectiveness and efficiency of the recheck policy.

The supplementary archive audit quantifies this cost limitation (Table 8). T1 records 802 observations and generation attempts, versus 647 for A0_NO_RECHECK, while their mean software episode times are 17.92 and 17.03 s. T1 also logs 165 search actions in addition to its 161 dedicated rerechecks. Across all seven archived policies, 42 of 3,360 episodes lack a state following the final ac-

tion; movement means therefore use the explicitly reported complete-trajectory subsets. These descriptive runtime and movement summaries do not constitute a matched flight-cost comparison. Rendering, perception, generation, and controller times cannot be separated from the available episode wall times.

7.5. Case study: Re-observation using real UAV imagery

The preceding experiments change observations by re-sampling orthographic xBD images. We constructed a separate exploratory case with real multi-elevation UAV images from MESSI [26] to ask whether a lower-altitude acquisition adds information beyond a repeat call or digital enlargement of the high image. During pilot preparation, the proposed parked-vehicle question did not provide stable positive regions in the three candidate evaluation sequences. We therefore fixed one static presence question before formal inference: whether a marked ground region contains visible tree or shrub canopy, supported by MESSI’s vegetation annotation (class 12) and visual review of both source images. This is a scene-content check, not disaster-damage recognition.

The case contains 12 fixed regions, four each from MESSI sequences 100_0002, 100_0003, and 100_0042, with six positive and six negative answers. Four regions from 100_0001 served only as pilot items and were excluded. For each evaluation sequence, we selected the frames nearest 100 and 30 m in the provided height relative to take-off: the observed high range was 99.9–100.2 m and the low range 29.8–30.3 m. We matched local image features across the two frames, estimated a homography, and visually checked the paired crops and labels before model outputs. These values are relative heights, not calibrated local above-ground distances. Frame selection by height defines an offline review; it does not reconstruct a flown descent decision.

Each item first presented Qwen2.5-VL-7B-Instruct with the high image and a red outline of the region. The resulting answer was copied into three isolated second-turn histories. Branch A repeated the same outlined high image. B was an 800×800-pixel digital enlargement of the corresponding crop from the approximately 100-m source image; it magnifies existing pixels and adds no new source detail. C was an 800×800-pixel crop from a separate image acquired at approximately 30 m. B and C show approximately the same ground footprint at matched display dimensions, but they are different source images. All branches used the same question, second prompt, greedy decoding, 16-token output limit, and one second answer call. Ground-truth masks and answers were withheld from prompts. All 12 items completed their planned first and three branch calls; **UNCERTAIN** and invalid answers counted as incorrect.

The primary paired comparison is C versus B (Table 9). C answered 11/12 correctly (91.67%), versus 10/12 (83.33%) for B, a descriptive difference of +8.33 percentage points. Relative to B, the new image made two incorrect answers correct and one correct answer incorrect. Relative

to the first high-image answer, C corrected five items and harmed one. A answered 5/12 (41.67%), versus 11/12 for C, but this secondary contrast also changes contextual extent. At the sequence level, B and C were tied at 3/4 in 100_0002 and 4/4 in 100_0003; C was 4/4 versus B’s 3/4 in 100_0042. Thus the aggregate difference is one answer among twelve and should not be read as a stable expected gain.

Figure 5 shows both directions of the paired outcome. In 100_0042_r16, the high crop produced **UNCERTAIN**, whereas the low image correctly rejected canopy on paving. In 100_0002_r02, the high crop correctly identified vegetation, whereas the low image incorrectly answered **NO** despite visible canopy at the left of the region. A new view can therefore clarify or lose task-relevant evidence. The sample is small and selected for clear cross-height labels; regions within a sequence are correlated. Registration and axis-aligned high cropping leave some footprint mismatch: the high crop’s bounding rectangle exceeds the mapped quadrilateral area by 2.8–15.9% across the twelve items. Changes in illumination, perspective, and scene context also accompany resolution changes. These fixed-observation branches measure the value of the supplied view; the following replay tests whether an adapted agent requests that view.

7.5.1. Action-gated offline replay

We next replayed the same twelve regions through DisasterClaw’s **AgentVqaController** and T1 task-conditioned utility gate. This exploratory follow-up was designed after the fixed-observation case. The offline environment initially exposed only the outlined high image; it released the paired low crop after an authorized descent action. The only candidate low frame and the approximately 100/30-m relative heights were fixed in advance. The agent arm had one reobservation action; the paired hold arm had zero. Both arms used the identical cached first Qwen response. The controller recorded each candidate answer, action, budget change, and stop reason. Reference answers were read only after the replay for scoring.

This is a MESSI-specific perception and decision adapter. The original T1 presence branch skips local descent, and MESSI supplies neither the disaster pre/post pair nor ChangeOS damage probabilities. Qwen instead supplied a YES/NO canopy answer and self-reported answer confidence; normalized binary entropy of that confidence replaced damage entropy. For the marked binary target, the adapter invoked the existing gate’s damage-target utility formula and its 0.5 uncertainty trigger, 0.05 flight-cost weight, and 0.05 minimum utility. It stipulated that the marked region was visible, used zero horizontal recentering, and bypassed the damage-specific **RecheckController.assess**. The controller’s answer, action-budget, trajectory, and stop paths remained in use with VLM answering and no search. This is an adapted offline decision test, not an unchanged T1 presence-policy or physical-flight test.

Table 8: Archived software work and geometric motion for the frozen benchmark (480 episodes per policy). Obs./calls gives both recorded observation count and generation-attempt count, which coincide here. Search and rechecks are logged actions; complete-motion denominators exclude terminal actions without a following state. Motion includes search and reobservation and is a geometric reconstruction, not physical flight. Wall time is measured software episode time; component timings and flight energy are unavailable.

Policy	Obs./calls	Search	Rechecks	Complete n	Horiz. (m)	Vert. (m)	Wall (s)
A0_NO_RECHECK	647	173	0	474	52.8	0.0	17.03
A1_RANDOM	1187	291	419	477	133.2	403.4	33.65
A2_ALWAYS	1706	319	923	464	122.1	897.3	51.07
A3U_RAW_ENTROPY	1379	278	626	475	159.4	576.0	39.61
A3_ENTROPY	1386	282	628	476	162.8	581.3	38.94
A4_CONFORMAL	1395	291	628	476	161.1	577.5	38.95
T1_TASK	802	165	161	476	59.6	165.8	17.92

Horizontal/vertical distances are means over complete trajectories; wall time is a mean over all 480 episodes. Action and observation columns are totals. Summed episode wall times are retained in the machine-readable audit; they are not elapsed parallel-job duration.

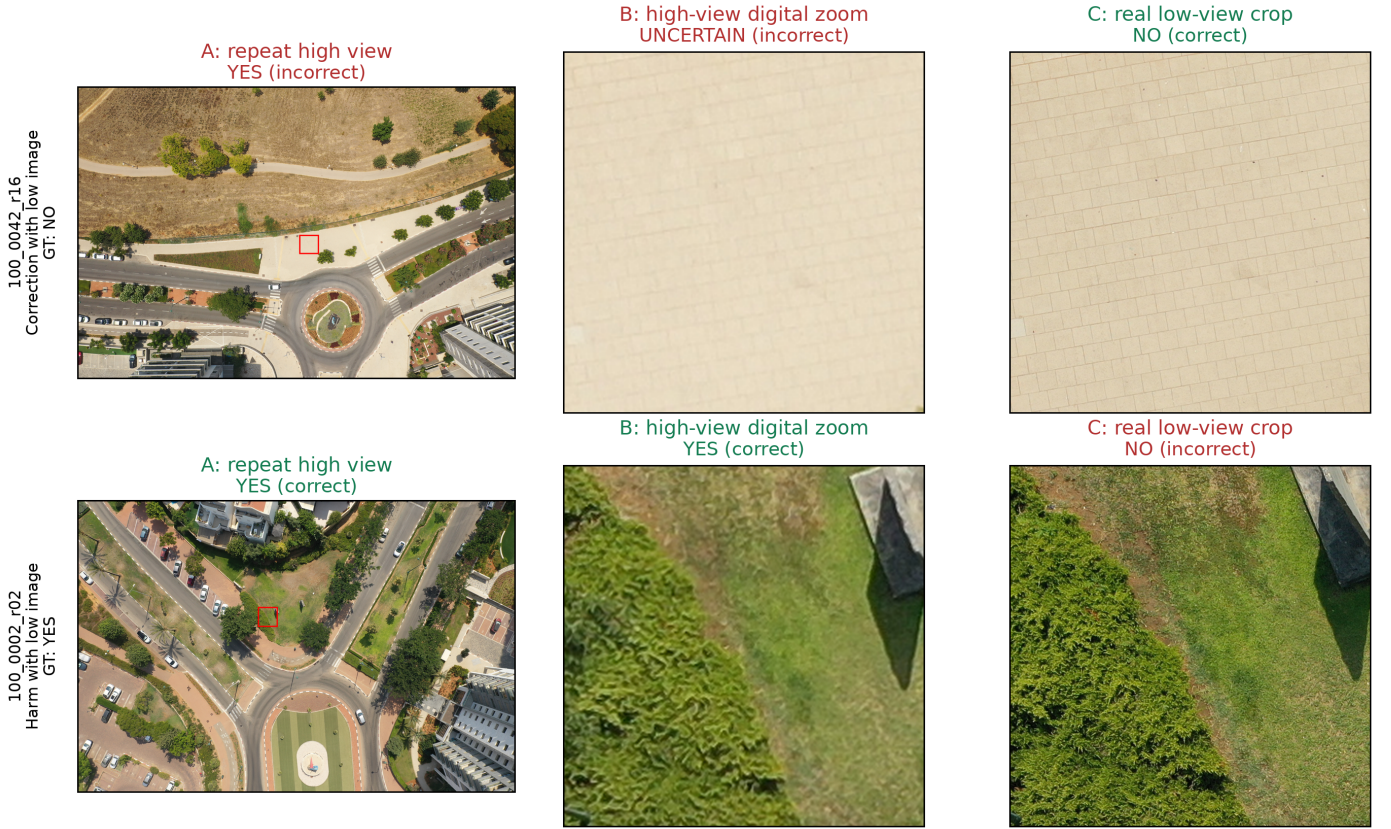


Figure 5: MESSI paired examples from the frozen case. A is the repeated high-altitude view with the target region marked. B is a digital enlargement of that region from the approximately 100-m image; C is the corresponding crop from a separate approximately 30-m image. B and C show approximately the same ground footprint at the same display size, but are not the same image: B enlarges existing high-view pixels, whereas C is a new lower-altitude observation. Frames were selected offline at 100.2 and 30.2 m relative altitude for 100_0042_r16 (top), and 99.9 and 30.3 m for 100_0002_r02 (bottom). Text gives second-turn model and reference answers (GT). The top row shows B-to-C correction; the bottom row shows B-to-C harm. Source images: MESSI [26], CC BY-NC-SA 3.0.

The hold arm answered 8/12 correctly, versus 10/12 for the action-enabled arm (Table 10); the two changed outcomes were corrections and none were harms. The agent requested the low image in all 12 episodes, and all episodes in both arms ended with `sufficient_evidence`. Thus the replay verifies that the adapted controller can trigger, acquire, incorporate, and stop after a real lower-altitude image, but the observed operating point offers no evidence

that it selectively suppresses unnecessary reobservations. Qwen’s reported high-view confidence was at most 0.75 in every item, which made every item exceed the entropy trigger. These descriptive results do not estimate a stable accuracy gain.

Table 9: Exploratory MESSI case on 12 fixed regions from three sequences (four per sequence; pilot excluded). A repeats the approximately 100-m image, B digitally enlarges its corresponding high-image crop, and C uses a crop from a separate approximately 30-m image. All use the same two-turn budget and first answer; B and C cover approximately the same ground footprint and are displayed at the same size. **UNCERTAIN** counts incorrect. Corrections and harms compare each final answer with the first answer.

Observation	Correct/ <i>n</i>	Acc. (%)	Corrected	Harmed	Uncertain
First high image	7/12	58.33	—	—	0
A: high repeated	5/12	41.67	0	2	2
B: high crop	10/12	83.33	4	1	2
C: real low crop	11/12	91.67	5	1	0

Table 10: Action-gated MESSI offline replay on the same 12 regions. The first model response is shared across arms. Rechecks count low-frame releases following a controller action.

Arm	Correct/ <i>n</i>	Accuracy (%)	Rechecks
No new observation	8/12	66.67	0
Agent autonomous recheck	10/12	83.33	12

7.6. Expanded evaluation on public MESSI pairs

We extended the case systematically after observing the 12-item results. The extension uses every annotation-consistent cell of a fixed 24-cell grid in five public MESSI descend sequences; 100_0001 remains a pilot sequence. This yields 107 regions from 120 evaluation-sequence cells, with 13 ambiguous or cross-height discordant cells excluded before the extension’s inference. We also paired each public 30-m frame in Agamim Paths A–C to its nearest 100-m frame by GPS and tested one fixed central region. Of 136 low frames, 120 passed distance and image-registration checks; 25 further pairs had ambiguous or discordant annotations, leaving 95 path pairs. The latter contain 83 positives and 12 negatives. The task is now *visible vegetation including grass, trees, and shrubs*, matching MESSI’s vegetation mask; it is distinct from the selected 12-item canopy task. Answers derive from agreement of high- and low-view masks, without visual confirmation of every region. The 202 units share frames and nearby ground content and must not be treated as independent flights.

The same high-repeat (A), high-source crop (B), and real low-image (C) branches completed all 202 vegetation items (Table 11). On descend regions C answered 83/107 correctly versus B’s 80/107, with seven C-over-B corrections and four harms. On path pairs C answered 83/95 versus B’s 86/95, with one correction and four harms. Aggregated descriptively, both B and C answered 166/202 correctly; their eight corrections and eight harms cancel. The selected 12-item positive difference therefore does not extend to a consistent low-image advantage under this broader public-pair protocol.

We also replayed the full extension through the self-reported-confidence adapter described above. The no-new-observation arm answered 138/202 correctly. The action-enabled arm answered 158/202, correcting 30 and harming

Table 11: Expanded, post hoc public MESSI vegetation evaluation. Entries A/B/C are correct counts. Help and harm compare C with B on the same items. Descend and path units use different systematic ROI rules and are correlated within geographic sequences.

Source	<i>n</i>	A	B	C	C/B help/harm
Descend grid	107	76	80	83	7/4
Path frames	95	79	86	83	1/4
Combined	202	155	166	166	8/8

10, but requested the lower image in every episode. This establishes a complete action/observation/answer trace across the expanded real-image set, while again offering no evidence of selective stopping from that uncertainty signal.

After diagnosing this all-recheck behavior, we ran a separate exploratory adapter that scores the Qwen next-token probabilities of YES and NO after a fixed binary prompt. Their normalized two-way entropy enters the same binary-target utility gate, with its uncertainty trigger set to 0.90 using four excluded pilot regions from 100_0001. This changes both the perception prompt and uncertainty measurement; the original damage-policy gate is still adapted to a marked vegetation target. The new adapter was frozen before its 202 evaluation episodes. Each hold/agent pair shares its first response, and only an approved recheck releases the predetermined low frame. The token scores are uncalibrated and should not be compared with self-reported-confidence accuracy as an isolated policy change.

The token-entropy adapter rechecked 102/202 items and stopped after the high view in 100/202 (Table 12). Its paired no-new-observation arm answered 155/202 correctly and the agent answered 165/202, with 26 corrections and 16 harms. The 100 stopped cases had 86 correct high-view answers; among the 102 rechecked cases, accuracy changed from 69/102 to 79/102. Descend regions gained seven net correct answers, while path pairs gained three; path B lost two and path C gained two. These post hoc results show selective requesting and stopping within the offline, preselected two-frame environment, but they do not establish the utility threshold’s calibration or a gain from physical flight.

Table 12: Exploratory selective MESSI replay with token-entropy uncertainty. Hold and agent share the first answer on each item; rechecks release the low frame. Corrected/harmed compare paired terminal answers.

Source	<i>n</i>	Hold	Agent	Rechecks	Corrected/harmed
Descend grid	107	83	90	57	17/10
Path frames	95	72	75	45	9/6
Combined	202	155	165	102	26/16

7.7. Full-release MESSI fixed observations and earlier Qwen replay

After the public-sample extension, we obtained the author’s full 2,525-image MESSI release and inventoried all image, annotation, and pose filenames. We applied the

same vegetation task and two-height selection rules to every available descend sequence and to the Agamim and IrYamim paths with both 100- and 30-m acquisitions. The previously used 100_0001 sequence remained pilot-only. One additional descend sequence failed the fixed 40-inlier registration criterion, and path pairs exceeding the prespecified 12-m GPS distance or failing annotation agreement were excluded. The Ha-Medinah Square path has only 60-m images, and the 50/70-m acquisitions do not enter this single-action protocol. The resulting 409 eligible regions comprise 280 descend-grid cells, 95 Agamim path pairs, and 34 IrYamim test-path pairs. The training paths replicate the earlier public-sample stratum; the newly available descend and test-path regions extend coverage. All selection and image hashes were fixed before the complete-release inference, with source and exclusion manifests retained separately from model inputs.

The original DisasterClaw presence gate explicitly answers from the wide view and never requests local descent. A diagnostic invoking the repository’s **AgentVqaController** and that unmodified gate on all 409 MESSI items made zero rechecks and answered 287/409 first views correctly. It still used a MESSI VLM first-view adapter rather than the complete ChangeOS/hybrid disaster pipeline. This zero-action result establishes that a positive recheck claim on the vegetation question requires a declared task adapter. Its answer accuracy is not directly comparable with the token-entropy adapter, which uses a different first-answer prompt and scoring method.

In the full-release fixed-observation experiment, high-source crop B and real low-image C each answered 341/409 correctly (Table 13); C corrected 13 B errors and harmed 13 B-correct answers. The descending sequences favored C by two answers, Agamim paths favored B by three, and IrYamim test paths favored C by one. Thus a real descent image did not consistently outperform digital enlargement across this eligible set.

Table 13: Fixed-observation results on all eligible full-release MESSI 100/30-m vegetation regions. A repeats the high image, B enlarges its source crop, and C uses the real low image. Help/harm compare C with B.

Source	<i>n</i>	A	B	C	C/B	help/harm
Descend grid	280	184	235	237		11/9
Agamim paths	95	79	86	83		1/4
IrYamim test path	34	24	20	21		1/0
Combined	409	287	341	341		13/13

In the earlier full-release Qwen token-entropy replay, the adapted controller requested 198/409 lower frames and stopped after the high view in 211/409 (Table 14). All 211 no-action gate outcomes cited uncertainty below the trigger; every episode ultimately stopped with **sufficient_evidence**. The paired hold arm answered 309/409 correctly versus 329/409 for action-enabled replay, comprising 44 corrections and 24 harms. The gain was 20 net answers on descend regions and three on Agamim paths;

on the previously unseen IrYamim test path, rechecks reduced correct answers from 25/34 to 22/34, with one correction and four harms. Every arm shared its first token score, and log audits found no low-frame read without an approved recheck; full action trajectories are retained. This was a separate perception adapter and is retained as exploratory context; the legacy-model experiment below is the direct evaluation of the existing DisasterClaw perception path.

Table 14: Task-adapted DisasterClaw action-gated offline replay on the full-release eligible regions. Hold forbids new observations; the agent uses the same first token score and may release one fixed low image. Corrected/harmed compare paired terminal answers.

Source	<i>n</i>	Hold	Agent	Rechecks	Corrected/harmed
Descend grid	280	212	232	139	34/14
Agamim paths	95	72	75	45	9/6
IrYamim test path	34	25	22	14	1/4
Combined	409	309	329	198	44/24

7.7.1. Single-temporal replay with DisasterClaw’s legacy perception models

The final MESSI controller evaluation uses the repository’s existing perception models and closed-loop controller on the same 409 eligible regions. The RescueNet YOLO checkpoint (six building-damage, vehicle, and water classes) has no vegetation category; we retain its detections as scene context and do not use them as vegetation evidence. Vegetation presence comes from the ADE20K SegFormer-B2 model, using tree (class 4), grass (9), plant (17), and palm (72). For a region, we map its mean soft vegetation occupancy f to the task score $p = \text{clip}[(f - 0.005)/(0.05 - 0.005), 0, 1]$ and answer YES at $p \geq 0.5$. The endpoints 0.005 and 0.05 are inherited from the earlier MESSI cohort-construction protocols: a negative region has at most 0.5% annotated vegetation at both heights, whereas a positive region has at least 5%. The adapter reuses these protocol boundaries as a linear heuristic, giving a YES threshold of $f = 0.0275$; they are not fitted calibration parameters. The mapping therefore incorporates prior knowledge of the benchmark selection rule, although individual annotation masks and reference answers are not supplied at runtime. These endpoints were already specified in the earlier descend/path protocols and carried into the single-temporal protocol; this is a documented design history, not an independently preregistered calibration study.

The model sees an unmarked registered ROI crop at each height. Each paired hold/agent run shares the exact high-view model output. The repository’s **AgentVqaController** retains its evidence memory, answer and stop path, action budget, and trajectory; a narrow single-temporal adapter supplies the vegetation evidence and deterministic presence answer. The **RecheckController** retains its normalized two-class entropy gate; the 0.50 threshold is inherited from its existing default and was not fitted on MESSI. It is

allowed one action. If it triggers, the offline environment releases the pre-registered 30-m crop and records a descend-only action matching the observed relative height difference. It does not execute horizontal recentering, search, or physical motion. The low image is not made available to either the perception callback or the controller before that action.

On the full 409-item set, the no-new-observation arm answered 369 correctly (90.2%), and the agent arm answered 372 correctly (91.0%). The controller requested 20 lower-view observations; six changed wrong answers to correct and three changed correct answers to wrong. By stratum, the agent changed 259/280 to 260/280 with 12 rechecks, 85/95 to 87/95 with seven, and 25/34 to 25/34 with one on the IrYamim test-path stratum (Table 15).

The 409-item hold arm makes only the first answer call. We therefore added a matched second-observation control for the exact 20 triggered items: after the shared high-view answer and identical controller trigger, one arm reprocessed the same high ROI while the other opened the registered low ROI. Each second-observation arm had the same controller state, one action, and two deterministic answer calls; only the image in the second call differed. The repeated-high arm answered 12/20 correctly, identical to its shared first answers, while the real-low arm answered 15/20 correctly (Table 16). Compared directly with repeated high, the new low view corrected six and harmed three; nine remained correct in both and two remained wrong in both. This isolates the observation change from the additional answer opportunity on the triggered subset. Across all 409 items the aggregate net change remains three answers, and the IrYamim path shows no accuracy gain. These item-level results are descriptive because regions share scenes and frames.

Table 15: Single-temporal DisasterClaw replay with the legacy YOLO and SegFormer models on all eligible MESSI regions. The hold and agent arms share the first high-view predictions. Rechecks count controller-authorized low-frame releases; corrected/harmed are paired terminal-answer changes.

Source	n	Hold	Agent	Recheck	C / H
Descend grid	280	259	260	12	3/2
Agamim paths	95	85	87	7	3/1
IrYamim test path	34	25	25	1	0/0
Combined	409	369	372	20	6/3

Table 17 follows one correction and one harm through both equal-budget branches. The controller issues the same descend request in each branch, with zero horizontal displacement; the repeated-high environment retains the original view, while the real-low environment opens the pre-registered lower frame. Full event payloads and controller records are provided in the machine-readable experiment trajectory file.

The stopping implementation first forms a deterministic YES/NO answer and assigns the default answer-path code `sufficient_evidence`. It then consults the recheck gate only while an action remains. After one recheck,

it returns the answer without another entropy test or a confidence-based abstention rule. Thus the near-boundary harmed example in Table 17 is a budget-constrained terminal answer, not evidence that uncertainty was resolved. We preserve the recorded code rather than relabeling the historical trace as `budget_exhausted`, which is a different controller exit path.

The 34 IrYamim regions belong to the MESSI Test split, but the same source regions had already appeared in earlier Qwen-only MESSI explorations. The RescueNet checkpoint, SegFormer checkpoint and class mapping, and inherited 0.50 trigger were fixed before this legacy-model replay; no IrYamim answer was used to tune them. We therefore report IrYamim as a test-path stratum but do not treat it as a data-naive untouched holdout.

Random selection under the same recheck budget. To assess whether the selected regions offer more value than an arbitrary allocation of the same observation budget, we added a supplementary uniform-random comparator (Table 18). Before supplementary inference, seeds 0–19 were fixed; each sampled 20 of the 409 eligible regions without replacement, using only region identifiers. The same archived high-view predictions, model checkpoints, vegetation-score mapping, controller evidence and answer paths, and registered lower frames were retained. Only the recheck selection rule changed. Both active policies therefore use 429 observations and deterministic answer evaluations per 409-region allocation, including exactly 20 lower views; this matches observation counts, not maneuvering time or energy. Unselected regions retain their first answer.

The 20 random allocations selected 265 distinct regions. Including the original entropy-triggered cohort gave 274 unique regions, each evaluated through a one-action controller episode on the same GPU. Lower crops were accessed only after the action callback. Deterministic per-region outcomes were reused across seeds, without cross-region evidence sharing; these allocations are not 20 independent model reruns. All 20 original triggered answers agreed with their archived outcomes in the same-device check.

Uniform random selection yielded a mean of 369.20 correct answers (sample standard deviation 0.95; range 367–371), compared with 372 for entropy selection and 369 for the initial high view. Random selection averaged 0.55 corrections and 0.35 harms, versus six and three for entropy selection. Thus entropy selection produced 2.80 more correct answers than the random mean, or 0.68 percentage points over the full cohort. None of the 20 sampled allocations reached 372. This descriptive comparison supports the value of the trigger’s allocation on the current cohort, while also showing that it selects opportunities for both correction and harm. Seed variation on these shared, spatially correlated regions is not a generalization confidence interval or evidence of a universally superior policy.

For the archived MESSI perception calls, the 409 initial high observations took 79.23 s in total (0.194 s per call), and

Table 16: Matched, equal-budget comparison on the 20 first-view-triggered regions. Both second-observation arms share the initial high-view output, controller trigger, one-action budget, and two answer calls; only the second image differs. Corrected and harmed compare real-low with repeat-high. Scores are uncalibrated.

Triggered set	n	Initial high correct	Repeat-high correct	Real-low correct	Corrected	Harmed	Unchanged: correct / wrong
All regions	20	12	12	15	6	3	9 / 2

Table 17: Representative complete decision traces from the matched triggered cohort. Both branches use two answer calls, one controller-authorized descend request with zero horizontal displacement, and retain the logged terminal code `sufficient_evidence`. After the single action, no further recheck is permitted; this code denotes the controller’s answer-return path, not a separate confidence test. Scores are rounded to three decimals.

Case / truth	High: score / answer	Gate: entropy / descent	Repeat-high: score / answer	Real-low: score / answer	Calls / terminal stop	Effect
100_0041_r11 / YES	0.218 / NO	$U = 0.757$; $\Delta z = -70.0$ m	0.218 / NO	0.809 / YES	2; sufficient evidence	Corrected
100_0004_r08 / NO	0.477 / NO	$U = 0.998$; $\Delta z = -70.1$ m	0.477 / NO	0.500 / YES	2; sufficient evidence	Harmed

Table 18: MESSI random-selection control with the same 20-recheck budget. Random values are means over all 20 frozen seeds; SD and range describe correct-answer counts across allocations. Each active allocation has 429 observations and deterministic answers.

Selection	Correct / 409	Corrected	Harmed
Entropy trigger	372	6	3
Uniform random	369.20	0.55	0.35

Random SD: 0.95; range: 367–371. Shared initial high-view score: 369/409. No seed was selected for reporting based on its outcome.

the 20 triggered low observations took 3.73 s (0.186 s per call). These recorded latencies cover the combined single-temporal perception path, including input preparation; component timings and physical flight time are unavailable. The agent episode timer excludes the reused first-view inference, so its elapsed time cannot be compared directly with the hold timer as an end-to-end policy speed measurement. The random comparator addresses allocation under a fixed observation count rather than computational throughput.

8. Limitations

The benchmark contains 44 ROIs from three events, and questions from nearby regions may share image context. Although development and final questions use distinct tiles and building UIDs, their ROIs and rendered footprints can overlap geographically. Together with the fixed ChangeOS checkpoint’s prior xBD exposure, these overlaps limit claims of spatial independence and transfer to unseen disasters. The final answer distribution is uneven, particularly for count questions. The image-aware pipelines substantially outperform the evaluated shortcut controls, but this result does not establish that every question requires visual evidence or exclude unmeasured shortcuts. Human review and deterministic annotation rules support reference construction without guaranteeing visual answerability for every

question.

The camera ladder resamples orthographic imagery and uses asymmetric pre-/post-disaster processing. Unlike 3D UAV simulators that model vehicle dynamics and onboard sensing, it cannot reproduce changes in viewing angle, occlusion, acquisition conditions, or motion dynamics during physical descent. The platform therefore evaluates an action-conditioned observation interface, not flight control, sensor-resource scheduling under bandwidth or energy limits, or vehicle safety. The spatial uncertainty score is conditional on a selected predicted target and does not fully capture uncertainty about which damaged building is nearest. The target switch in Fig. 4 illustrates this unresolved source of harm.

The initial MESSI case contains twelve manually selected canopy-presence regions and a one-answer C-versus-B advantage. The later vegetation extension includes grass and has a different question, so the cohorts are not pooled. Although its source is the complete 2,525-image release, only 409 regions meet the fixed 100/30-m pairing, registration, and cross-height mask-agreement rules. Thus the scored units are not all source images or independent flights. Excluding ambiguous masks and distant or unregistered path pairs can favor clear cases. Agamim paths are strongly positive-skewed, and nearby items share frames or ground content. Only 34 IrYamim test-path regions qualify, from one geographic path. Frame pairs and candidate low views are fixed offline; planar registration and the high-image bounding crop are approximate, and relative altitude is not local above-ground clearance. The real low view ties the enlarged high crop overall, with 13 corrections and 13 harms.

The full-release controller replay uses a declared single-temporal task adapter. The existing YOLO has no vegetation class and contributes context detections only; SegFormer ADE20K class probabilities supply vegetation occupancy. The occupancy-to-answer mapping and its binary

entropy are uncalibrated, and the vegetation classes are not specialized for MESSI UAV imagery. The adapter replaces the disaster-damage perception and hybrid answer path with current-frame vegetation evidence and a deterministic presence answer; it does not run ChangeOS or a generative VLM. The repository’s AgentVqaController evidence, budget, action, and stop loop and the RecheckController threshold state machine remain in use, with a generic vegetation evidence label added. Each lower view is fixed and registered in advance, and the replay records a descend-only action without executing flight or recentering. Across 409 items, the action-enabled arm gains three correct answers overall (six corrections and three harms). For the 20 triggered cases, the matched repeat-high control receives the same second-call budget as the real-low arm: repeat-high scores 12/20 and real-low 15/20, with nine correct and two wrong in both branches. The 34 IrYamim Test regions were previously included in Qwen-only exploratory runs; the legacy-model checkpoint, class mapping, and inherited 0.50 trigger were fixed before this replay, and no results from these 34 regions were used to tune them. We therefore do not treat IrYamim as an untouched holdout, and do not tune further on these items. These results establish an exercised offline decision loop on real multi-height imagery, not disaster-damage performance, autonomous view selection, physical flight feasibility, or a reliable benefit from descent.

Earlier full-release experiments used Qwen-only perception adapters and different decision signals, including a post hoc token-entropy threshold selected on pilot items. They remain exploratory and are reported separately from the legacy-model replay; their absolute accuracies cannot be compared as a controlled policy contrast because the observation interpretation and answer module differ. The fixed-observation B/C comparison also uses Qwen and should not be pooled with the YOLO/SegFormer controller outcomes.

A0_NO_RECHECK permits search, and policies may differ in subsequent answer-call counts. The matched-call diagnostic controls for an additional answer opportunity at recheck transitions but does not follow both branches to episode termination. Perception may remain nondeterministic under identical image pixels, and the 95% ROI-cluster interval for the matched-call difference includes zero. The old-view agreement check varies prompts and seeds, so it does not exclude model variance under strictly identical calls. The qualitative examples illustrate observed mechanisms without establishing a generally positive recheck effect.

Recheck counts exclude search and do not capture total rendering, perception, generation, or flight costs. Of the 3,360 episodes, 42 lack a recorded state after the terminal fly action. This omission affects complete-motion summaries but leaves the terminal-answer denominator unchanged.

Accordingly, the reported gains concern answer-level observation value, not a measured trade-off against flight time or energy. Software inference latency, where recorded,

cannot be interpreted as aircraft maneuvering time. In the MESSI adapter, the terminal `sufficient_evidence` code does not certify resolved uncertainty once the action budget has been consumed.

The random-budget comparator is a post hoc supplement on the same 409 regions, with the 20 seeds fixed before its own inference. It compares allocation at a fixed observation count using shared first predictions and reused deterministic per-region outcomes. Its seed variation is conditional on this cohort, not independent replication across scenes, and neither random nor entropy selection is matched on physical motion cost. The observation-count and software-time audit does not recover separate rendering, perception, generation, and controller costs for the main benchmark.

9. Conclusion

DisasterClaw contributes an evaluation component for embodied UAV intelligence: it keeps the geographic scope and reference answer fixed while the agent’s action changes the observation. Its episode records connect actions, predicted building evidence, answer calls, and terminal outcomes. The four task families capture a trade-off between local detail and regional coverage that building-classification accuracy alone does not describe.

On the frozen final set, image-aware pipelines outperform the evaluated text-only and permitted-metadata controls. At matched recheck transitions, changed observations yield more corrections than harms, but the ROI-cluster interval for the average difference includes zero. The task-wise point estimates are positive for damage and count and negative for spatial questions. T1_TASK achieves the highest observed complete-policy accuracy, although its adjusted contrast with the search-enabled A0_NO_RECHECK policy also includes zero. In the full MESSI release, existing single-temporal YOLO and SegFormer models were connected to DisasterClaw’s evidence and recheck controllers for 409 vegetation regions. The action-enabled arm used 20 lower views and answered 372/409 correctly, compared with 369/409 for paired hold; six answers were corrected and three harmed. On the same 20 triggered cases, an equal-budget repeat-high arm answered 12/20 correctly and the real-low arm answered 15/20, isolating the second-view contribution from the additional answer opportunity. A supplementary random-selection control with the same 20-recheck budget averages 369.20/409 over 20 fixed seeds (range 367–371), below the entropy trigger’s 372/409. This supports selective allocation on the evaluated cohort; the seed spread does not establish cross-scene generalization. The 34 IrYamim Test regions had appeared in earlier Qwen-only explorations and are not an untouched holdout; both current arms answer 25/34 correctly. These results show that an adapted controller can gate access to new observations in a real-image offline replay. The net gain is small, and physical flight, vehicle dynamics, and transfer of the unchanged disaster-damage pipeline remain untested.

Declarations

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CRedit author statement. *To be assigned and approved by the named authors.*

Declaration of competing interest. *To be confirmed by all authors.*

Data and code availability. We will release the DisasterClaw code, benchmark task definitions, frozen protocols, audited episode traces, analysis scripts, and a runnable minimal reproduction package with its example xBD tile and labels on GitHub. The repository will provide environment instructions, file checksums, and instructions for obtaining the full xBD imagery and model checkpoints from their respective sources. The planned release repository is <https://github.com/emmmmmnnbbb/DisasterClaw>. This availability statement describes the intended release; completeness of the public archive remains to be verified before submission.

Generative AI assistance. Generative AI tools were used to assist manuscript restructuring, English-language editing, reference checking, and preparation of analysis and typesetting scripts. The authors are responsible for the scientific claims, experimental design, reported results, and final manuscript content. *Author confirmation of this disclosure and final review is required before submission.*

Appendix A. Protocol and provenance

The frozen execution protocol identifies the final task set and consolidated human-review report, three-event manifest, ChangeOS checkpoint, source fingerprint, prompt, policy configuration, and analysis scripts by SHA-256. The planned GitHub release will include the protocol, task and scene manifests, dependency specifications, replay instructions, and verified links to the original imagery and checkpoint sources.

The final run consists of 12 shards with 3,360 question-policy-repeat rows. The shard audit reports complete question and configuration coverage and zero execution errors. This checks execution consistency, not independent reconstruction of the full environment. A separate cost audit finds that 42 episodes end with a fly action and no subsequent recorded state; complete-motion summaries therefore use 3,318 episodes. The terminal-answer analysis retains all 3,360. Search-action detail and component-level render, perception, controller, and Qwen costs are not available, so the action-cost comparison is restricted to executed dedicated rechecks and observed episode wall time.

The supplementary-control report aggregates 12 matched-view shards (161 valid forks, no missing pairs or execution errors) and four shortcut shards (960 valid

text-only or metadata-only predictions, no call errors or invalid outputs). The planned GitHub release will include the machine-readable report, task-wise counts, ROI-cluster intervals, and protocol notes describing the fixed-view branch, allowed shortcut fields, and diagnostic boundaries. It will also include the matched-view and shortcut-control runners and the report-generation script so these results can be regenerated from the released inputs.

For the camera-ladder diagnostic, the construction boundary is summarized below. The table is a reproducibility description, not an ablation of preprocessing choices.

Table A.1: Camera-ladder preprocessing construction used by the diagnostic.

Stage	Pre-disaster	Post-disaster
Geographic crop	matched window	current footprint
Scale handling	assumed source scale	footprint-dependent sampling
Alignment	paired affine placement	paired affine placement
Detector input	retained source-scale channel	resized current view

Appendix B. Minimal frozen episode reproduction

The planned GitHub release will include a minimal example that replays one frozen T1_TASK damage episode, `hurricane-michael_00000435_post_disaster__damage_0001_9430`, generation repeat 0. The example contains the paired 1024×1024 xBD tile images, the post-disaster polygon/UID annotations, a scene manifest with geographic ROI and pixel-to-geographic transform, a compact record of the episode’s observations and actions, a standard-library Python replay script, and exact expected output. SHA-256 digests in the manifest detect asset substitution. This example can be run without detector or language-model weights; it reproduces scene linkage, observation-window geometry, the archived action trace, and reference-answer scoring, but does not rerun perception or generation.

The environment for this replay is Python 3.11 with no third-party packages. The GitHub README will specify how to run the replay script and compare its output with the expected result. For fresh model inference, the release will provide full environment specifications, the frozen task and review manifests, and instructions for obtaining the ChangeOS and Qwen2.5-VL-7B-Instruct checkpoints. A fresh inference run need not reproduce the archived candidate sequence exactly.

The replay first verifies the two image hashes and their dimensions, locates the marked building by xBD UID in the post-disaster label, and computes its binary damage reference from the original `major-damage` subtype. It then constructs each observation footprint from the recorded pose using the 60° field of view and the same geographic conversion used by the renderer. Table B.1 gives the compact

recorded action sequence. The final **Report** answer is compared with the annotation-derived reference, yielding the expected output **reference=damaged**, **answer=damaged**, and **correct=true** (the underlying task vocabulary is Chinese). The example identifies its source frozen record by question ID, generation repeat, and shard.

Table B.1: Expected trace of the single frozen damage episode. Candidate answers are shown in English for readability; the replay emits the original Chinese labels.

Step	Height (m)	Candidate	Action
0	1330.22	no damage	Fly/reobserve
1	886.82	no damage	Fly/reobserve
2	443.42	damaged	Report

The GitHub release will also include a supplementary audit script and its machine-readable output. The script reads the matched-call shards, development camera-ladder report, both frozen question manifests, xBD labels, and final T1 traces. Its output contains the schedule-exclusion calculation, old-view answer agreement, building-UID and geographic overlap audit, and the TP/FP/FN counts in Table 5. No new model inference is needed for these audit calculations.

Appendix B.1. MESSI archived scoring and fresh inference

The existing MESSI archive has separate entry points for rescored recorded outcomes and rerunning perception. From the repository root, `python3` followed by `experiments/messi_case/scripts/score_single_temporal_agent.py` recomputes the three-stratum hold/agent summary. The corresponding `experiments/messi_case/scripts/score_trigger_repeat_control.py` checks the triggered IDs and shared initial predictions and recomputes the equal-budget repeat-high versus real-low comparison. These scripts read the archived episodes and separate reference manifests; they do not rerun the models or validate fresh-inference reproducibility.

Fresh inference uses `experiments/messi_case/scripts/run_single_temporal_agent.py` with `-split` set separately to `full_descend`, `full_train_paths`, and `full_test_paths`, and `-device` selecting the available device. The matched repeat-high runs additionally use `-triggered-repeat-high`. Input preparation, model requirements, and run ordering are documented in `experiments/messi_case/single_temporal_agent_protocol.md`. This route requires the prepared MESSI crops and model weights; the archived scoring route requires the episode logs and reference manifests. Local entry points do not establish completeness of the planned public release.

The random-budget supplement is specified in `experiments/messi_case/supplement_random_budget/protocol.md`; its frozen selection manifest defines all 20 seeds. GPU traces, model provenance, and per-seed results are stored under `experiments/messi_case/supplement_random_budget_gpu/`. The entry `experiments/messi_case/scripts/score_random_budget_supplement.py` recomputes

the comparison from these records without inference. Fresh runs use `experiments/messi_case/scripts/run_random_budget_gpu.sh` with a device argument such as `cuda:0`. The separate `scripts/benchmarks/report_supplement_costs.py` audits the frozen benchmark calls, geometric motion, and software wall times reported in Table 8.

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